Physical Chemistry (I)

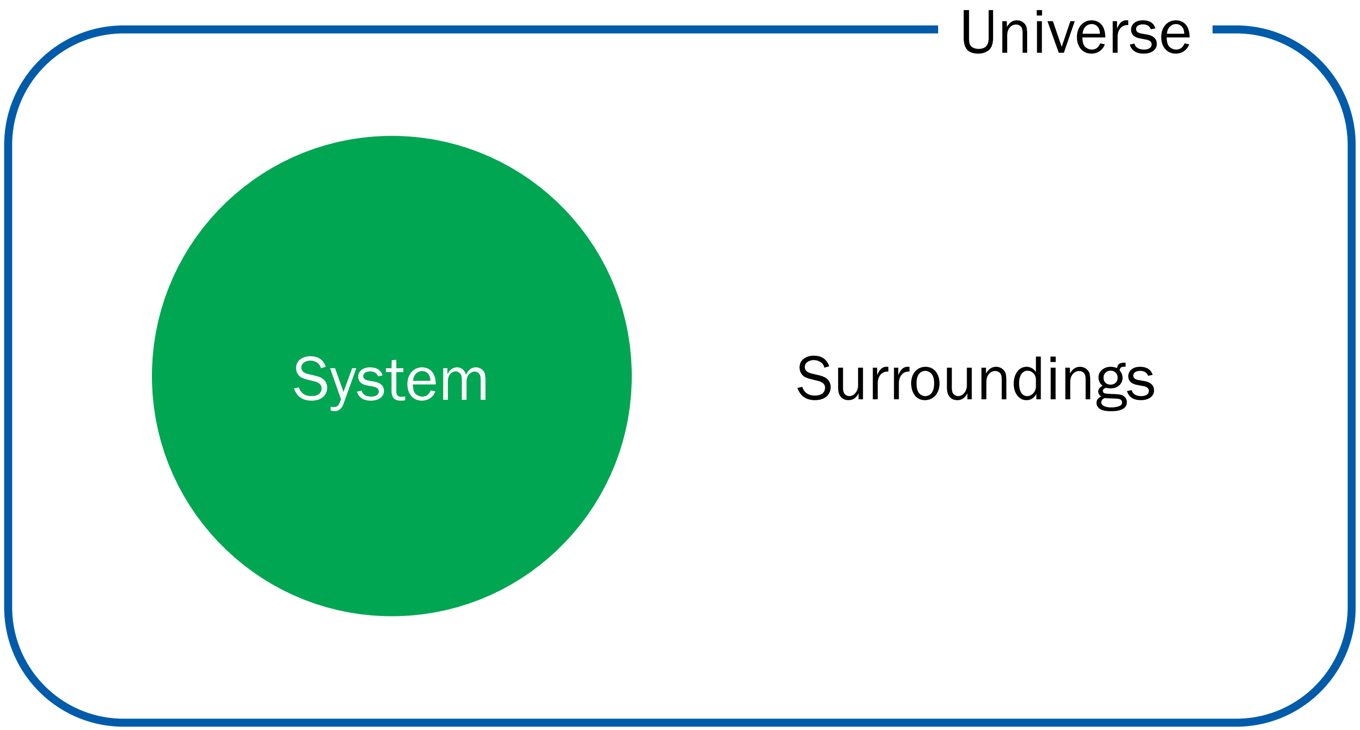
Lecture 03

Kinetic Molecular Theory

In the Last Class

- Basic Concepts
  - System: properties, equilibrium
  - System state and state change
- Gases
  - Why gases?
  - Perfect gas
  - Kinetic molecular theory

System and Surroundings



System Properties

- A system can have different physical properties
- A single value will be assigned for each property

- Useful classification
  - Extensive properties: $f(\mathrm{A}) + f(\mathrm{B}) = f(\mathrm{A} + \mathrm{B})$
  - Intensive properties: $f(\mathrm{A}) + f(\mathrm{B}) \neq f(\mathrm{A} + \mathrm{B})$

Equilibrium

- An isolated system is in equilibrium when its properties remain constant with time

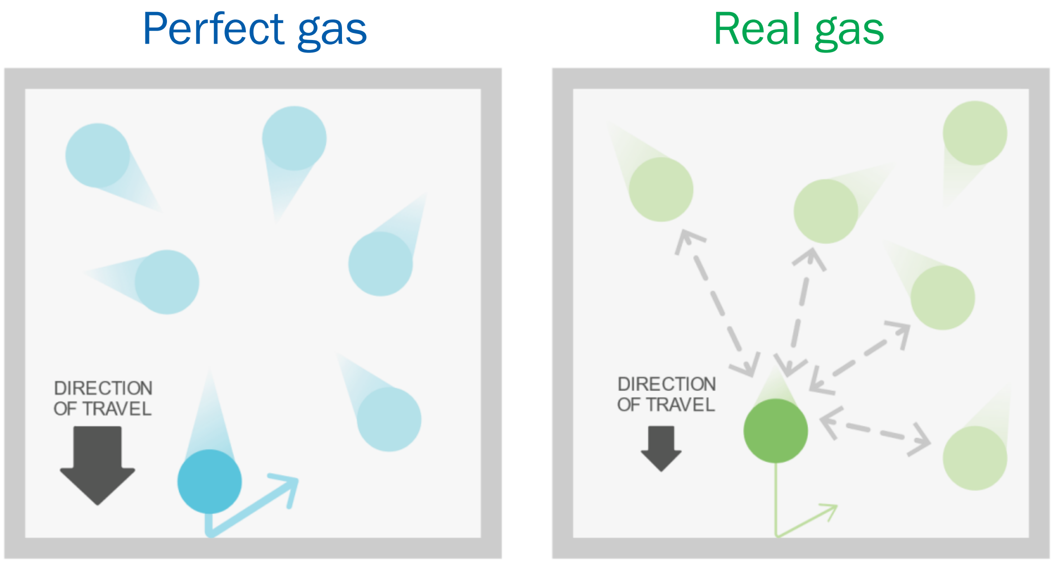
- Mechanical equilibrium: constant p and V
- Diffusive equilibrium: constant n
- Thermal equilibrium: constant T

State of a System

- The state of a system is defined by its properties
- For a pure substance: (n, V, p, T)
- State change: $(n_1, V_1, p_1, T_1) \rightarrow (n_2, V_2, p_2, T_2)$

Perfect Gas

- Perfect gas (ideal gas): no interactions
- Real gas: finite interactions between molecules



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Equation of State

- The state of a pure gas is defined by (n, V, p, T)

- Equation of state: relationship of the 4 variables

$$p = f(T, V, n)$$

- Equation of state of a perfect gas: perfect gas law

$$p = n\underbrace{R}_{\text{gas constant}}T/V$$

Kinetic Molecular Theory

Total energy = Σ (kinetic energies of molecules)

1. Molecules with mass m in ceaseless random motion
2. Molecules with negligible size
3. Elastic collisions: the total kinetic energy is conserved

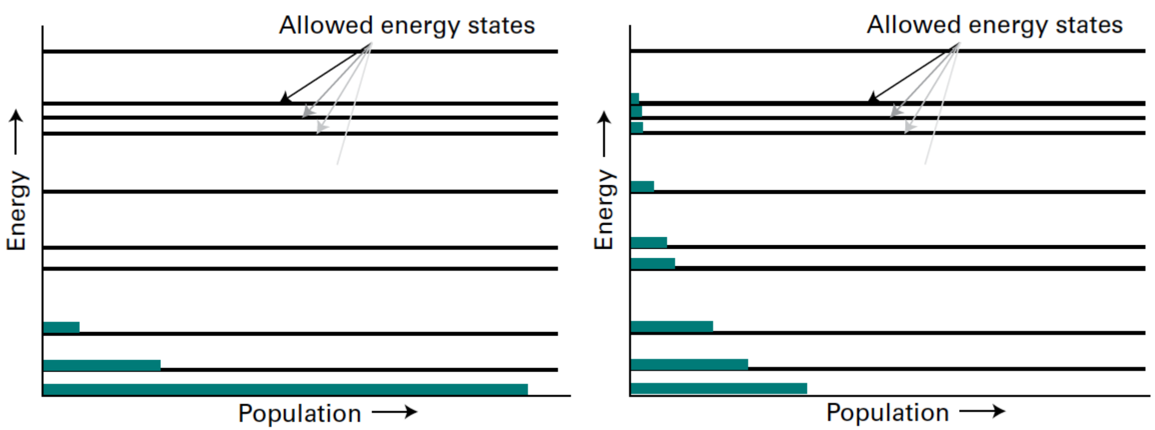
Last Class

Boltzmann Distribution

Boltzmann constant

Probability $\propto e^{-E/k_{\mathrm{B}}T}$

(at thermal equilibrium)



(a) Low temperature (b) High temperature

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Prologue Figure 2)

Boltzmann Distribution

- Probability density for energy E

$$f(E) \propto e^{-E/k_{\mathrm{B}}T}$$

$$f(E) = Ke^{-E/k_{\mathrm{B}}T}$$

- $f(E)\mathrm{d}E$: the probability that a molecule has an energy in the range $[E, E + \mathrm{d}E]$

Velocity Distribution

$$f(E) = Ke^{-E/k_{\mathrm{B}}T}$$

As $E = mv^2/2 = mv_x^2/2 + mv_y^2/2 + mv_z^2/2$,

$$f(v_x, v_y, v_z) = Ke^{-mv_x^2/2k_{\mathrm{B}}T}e^{-mv_y^2/2k_{\mathrm{B}}T}e^{-mv_z^2/2k_{\mathrm{B}}T}$$

After normalization,

$$f(v_x, v_y, v_z) = \left(\frac{m}{2\pi k_{\mathrm{B}}T}\right)^{3/2} e^{-m(v_x^2+v_y^2+v_z^2)/2k_{\mathrm{B}}T}$$

Velocity to Speed

- $f(v_x, v_y, v_z)\mathrm{d}v_x\mathrm{d}v_y\mathrm{d}v_z$:

Probability that a molecule has a velocity in the range

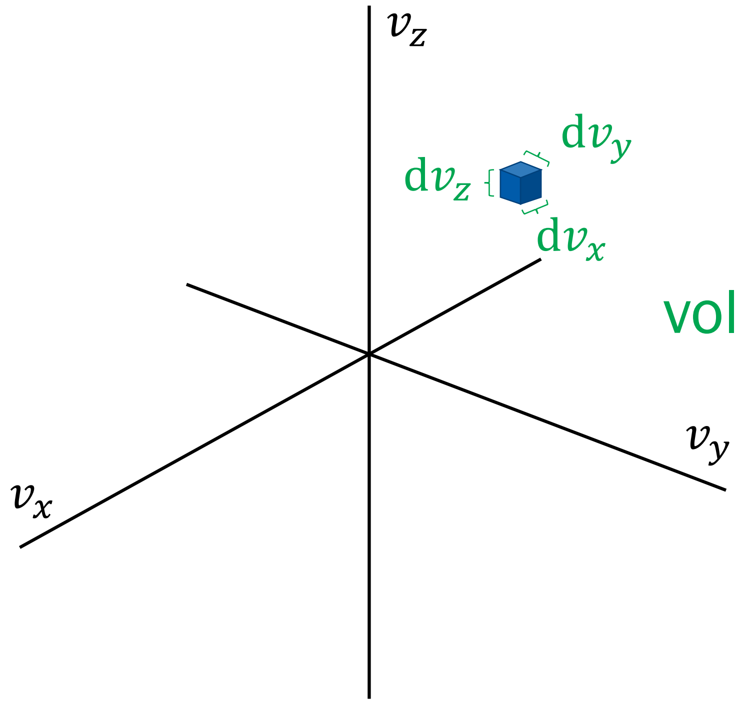
$$[v_x, v_x + \mathrm{d}v_x], [v_y, v_y + \mathrm{d}v_y], [v_z, v_z + \mathrm{d}v_z]$$

- $f(v)\mathrm{d}v$:

Probability that a molecule has a speed in the range $[v, v + \mathrm{d}v]$

Can we obtain $f(v)\mathrm{d}v$ from $f(v_x, v_y, v_z)\mathrm{d}v_x\mathrm{d}v_y\mathrm{d}v_z$?

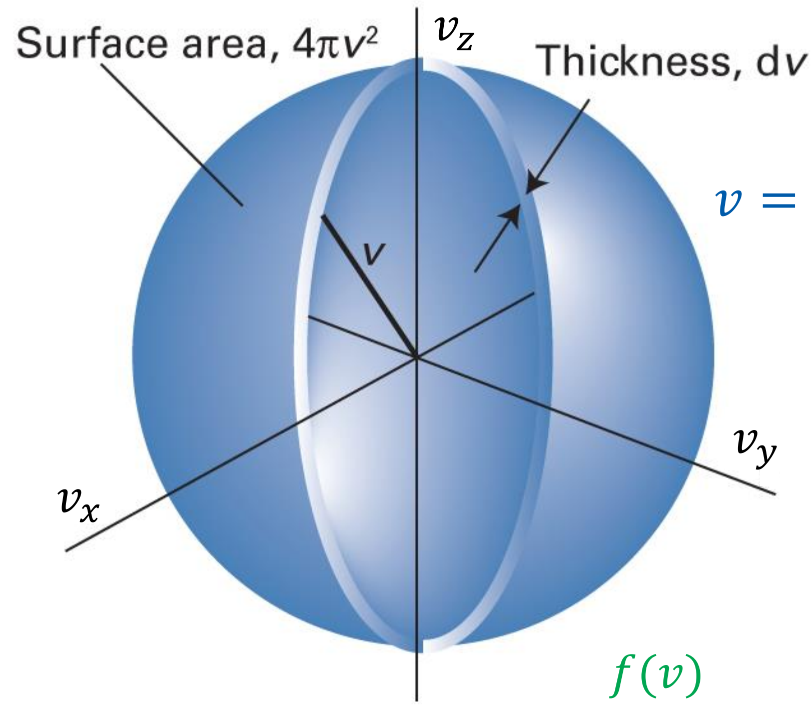
Velocity Space



volume: $\mathrm{d}v_x \mathrm{d}v_y \mathrm{d}v_z$

$f(v_x, v_y, v_z)\, \mathrm{d}v_x \mathrm{d}v_y \mathrm{d}v_z$ = function × volume

Velocity Space



$$v = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

$$\text{function} \times \text{volume} = \overbrace{f(v_x, v_y, v_z) \times 4\pi v^2}^{f(v)} \mathrm{d}v$$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 1B.3)

Topic 1B

Maxwell-Boltzmann Distribution

$$f(v) = f(v_x, v_y, v_z) 4\pi v^2$$

$$f(v) = 4\pi \left(\frac{m}{2\pi k_{\mathrm{B}} T}\right)^{3/2} v^2 e^{-m\left(v_x^2+v_y^2+v_z^2\right)/2k_{\mathrm{B}}T}$$

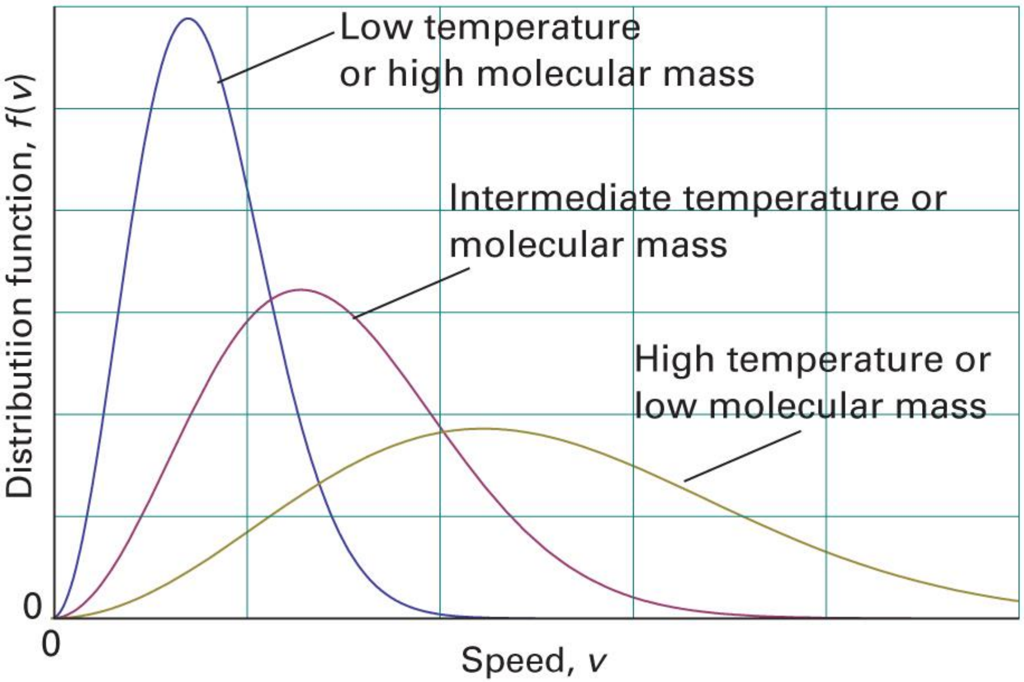
$$f(v) = 4\pi \left(\frac{m}{2\pi k_{\mathrm{B}} T}\right)^{3/2} v^2 e^{-mv^2/2k_{\mathrm{B}}T}$$

$R = k_{\mathrm{B}} N_{\mathrm{A}}$
$M = m N_{\mathrm{A}}$

$$f(v) = 4\pi \left(\frac{M}{2\pi RT}\right)^{3/2} v^2 e^{-Mv^2/2RT}$$

Interpretations

$$f(v) = 4\pi\left(\frac{M}{2\pi RT}\right)^{3/2} v^2 e^{-Mv^2/2RT}$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 1B.4)

Characteristic Speeds

1. Mean speed $\langle v \rangle = v_{\mathrm{mean}}$

$$\langle v \rangle = \int_0^\infty v f(v)\,\mathrm{d}v$$

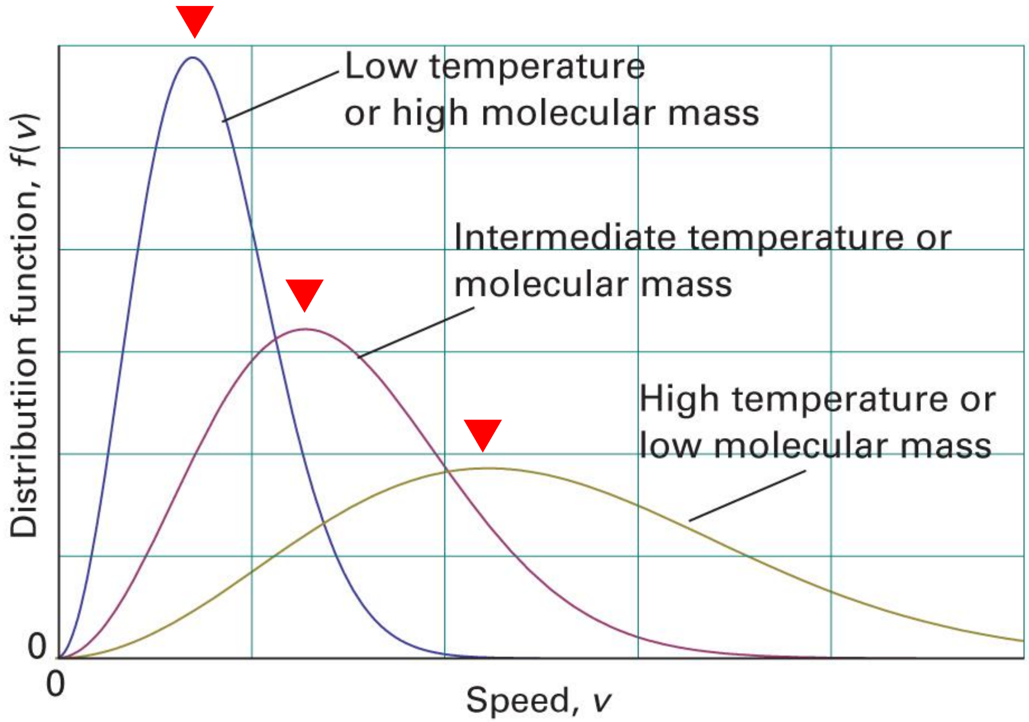
2. Root-mean-square speed v_{rms}

$$v_{\mathrm{rms}} = \sqrt{\langle v^2 \rangle}$$

$$\langle v^2 \rangle = \int_0^\infty v^2 f(v)\,\mathrm{d}v$$

Characteristic Speeds

3. Most probable speed v_{mp}



$$\frac{\mathrm{d}f(v)}{\mathrm{d}v} = 0$$

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figure 1B.4)

Characteristic Speeds: KMT

1. Mean speed

$$\langle v \rangle = \left(\frac{8RT}{\pi M}\right)^{1/2}$$

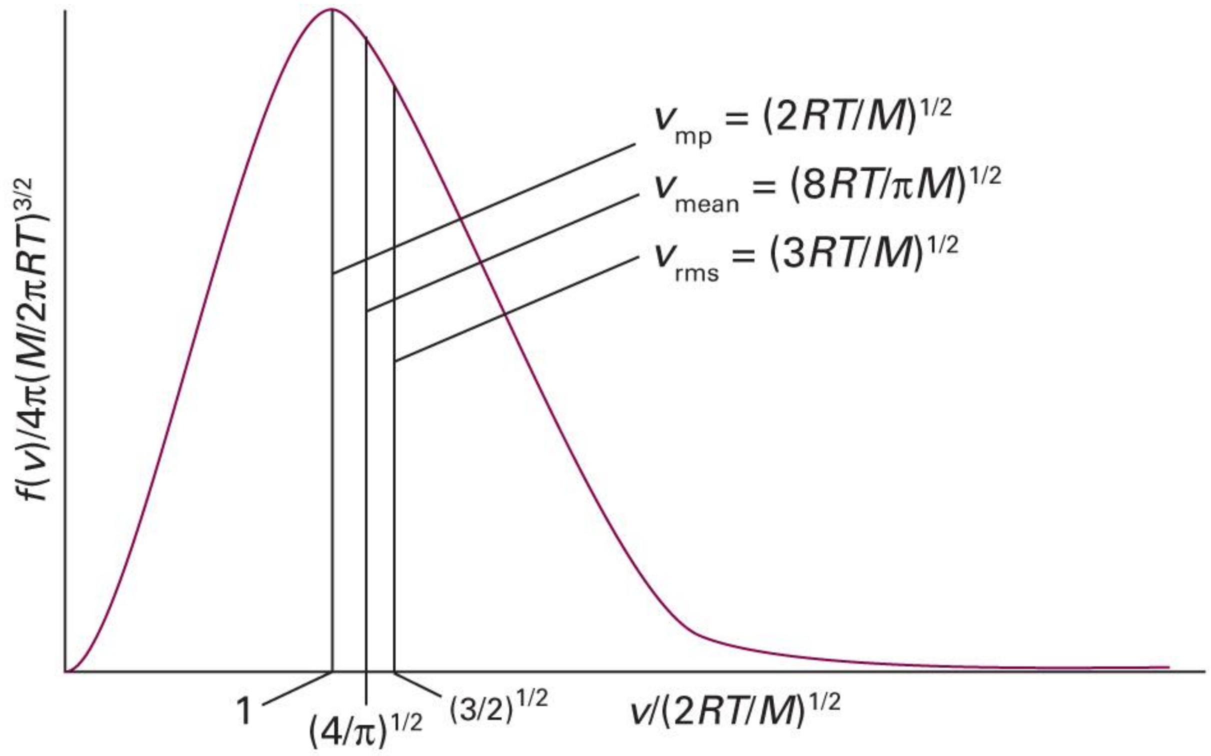
2. Root-mean-square speed

$$v_{\mathrm{rms}} = \left(\frac{3RT}{M}\right)^{1/2}$$

3. Most probable speed

$$v_{\mathrm{mp}} = \left(\frac{2RT}{M}\right)^{1/2}$$

Characteristic Speeds: KMT



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figure 1B.6)

Example 1B.1

Calculate v_{rms} and the mean speed, $\langle v \rangle$, of N_2 molecules at 25 °C.

For N_2, M = 28.02 g mol^{-1} = 0.02802 kg mol^{-1}

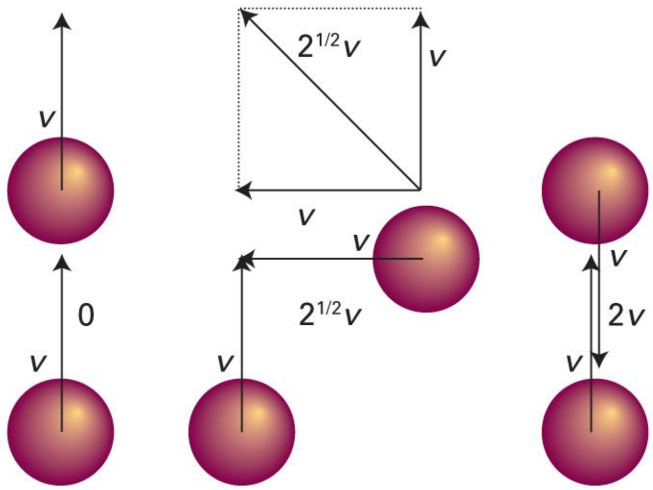
T = (273 + 25) K = 298 K

$$v_{\mathrm{rms}} = \left(\frac{3RT}{M}\right)^{1/2} = \left\{\frac{3 \times (8.3145\ \mathrm{J\ K^{-1}\ mol^{-1}}) \times (298\ \mathrm{K})}{0.02802\ \mathrm{kg\ mol^{-1}}}\right\}^{1/2} = 515\ \mathrm{m\ s^{-1}}$$

$$\langle v \rangle = \left(\frac{8RT}{\pi M}\right)^{1/2} = \left\{\frac{8 \times (8.3145\ \mathrm{J\ K^{-1}\ mol^{-1}}) \times (298\ \mathrm{K})}{3.1416 \times 0.02802\ \mathrm{kg\ mol^{-1}}}\right\}^{1/2} = 475\ \mathrm{m\ s^{-1}}$$

Mean Relative Speed

"Mean speed with which a molecule approaches another"



$$v_{\mathrm{rel}} = 2^{1/2}\langle v \rangle = \left(\frac{16RT}{\pi M}\right)^{1/2}$$

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figure 1B.7)

Mean Relative Speed

For two molecules of different types (A and B),

$$v_{\mathrm{rel}} = \left(\frac{8k_{\mathrm{B}}T}{\pi\mu}\right)^{1/2}$$

where reduced mass $\mu = m_{\mathrm{A}}m_{\mathrm{B}}/(m_{\mathrm{A}} + m_{\mathrm{B}})$

If $m_{\mathrm{A}} = m_{\mathrm{B}}$,

$$\mu = m_{\mathrm{A}}^2/(2m_{\mathrm{A}}) = m_{\mathrm{A}}/2$$

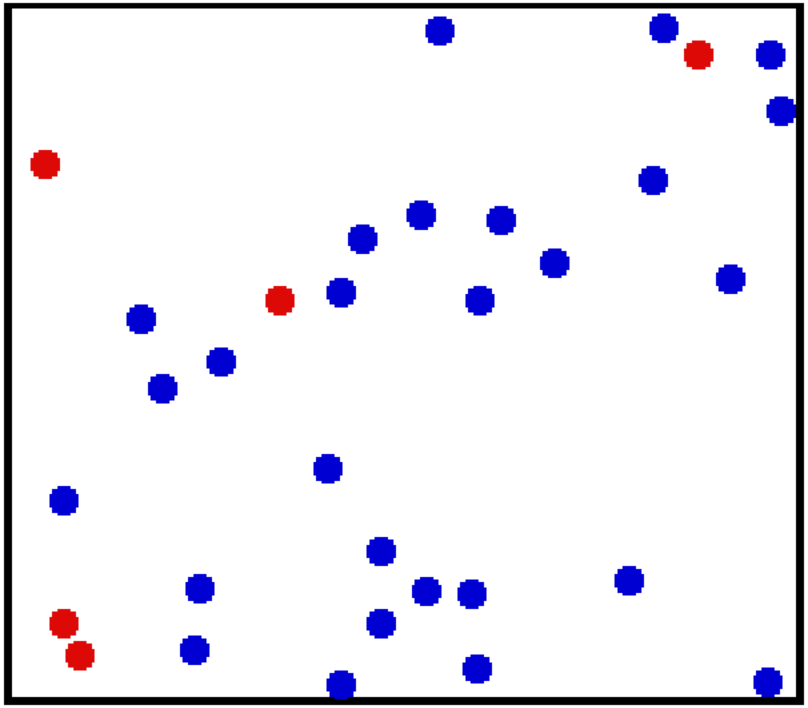
$$v_{\mathrm{rel}} = \left(\frac{16k_{\mathrm{B}}T}{\pi m}\right)^{1/2} = \left(\frac{16RT}{\pi M}\right)^{1/2}$$

Brief Illustration 1B.1(a)

The mean speed of N_2 molecules at 25 °C is 475 m s^{-1}. What is their relative mean speed?

$$v_{\mathrm{rel}} = 2^{1/2}\langle v\rangle = \sqrt{2} \times (475\ \mathrm{m\ s^{-1}}) = 671\ \mathrm{m\ s^{-1}}$$

Application: Collisions



(Image: https://commons.wikimedia.org/wiki/File:Translational_motion.gif)

Consequences of Collisions

- Molecule-molecule collisions

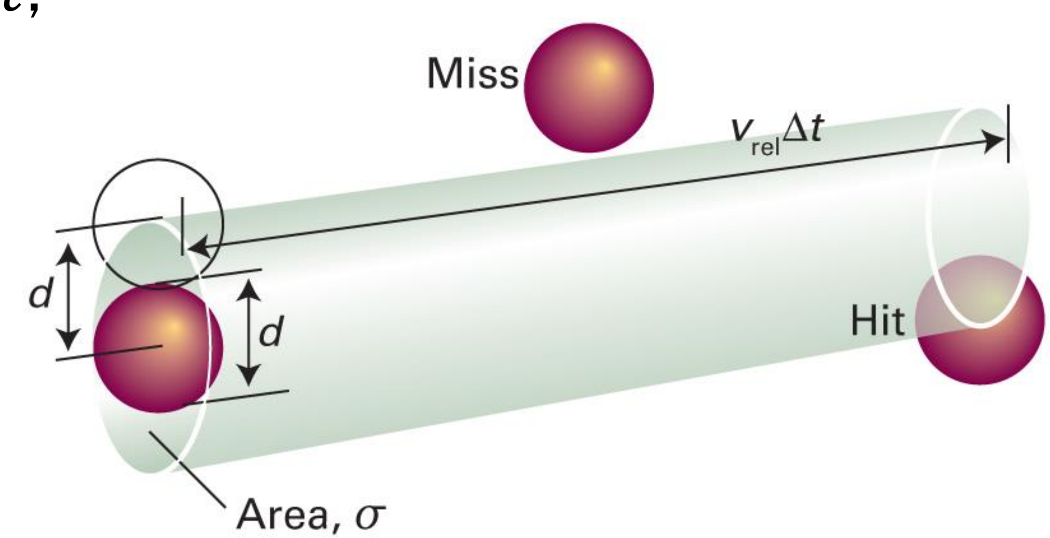
Molecular origin of chemical reactions

- Molecule-wall collisions

Molecular origin of pressure

Molecule-Molecule Collisions

For time Δt,

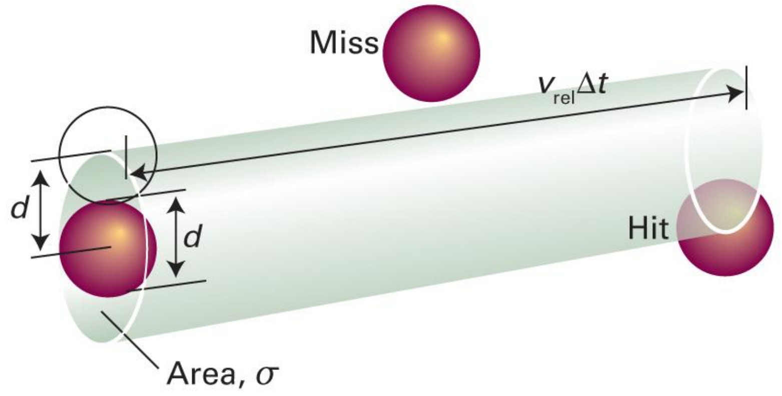


d = collision diameter

$\sigma = \pi d^2$= cross-sectional area

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 1B.8)

Collision Frequency



- Volume of the collision tube = $\sigma \times v_{\text{rel}}\Delta t$
- Number of molecules in the tube = volume × density

  $= (\sigma v_{\text{rel}}\Delta t) \times (nN_{\text{A}}/V)$
- Collision frequency $z = \sigma v_{\text{rel}} nN_{\text{A}}/V$

$\boldsymbol{p}$-dependences of $\boldsymbol{z}$

$$z = \frac{\sigma v_{\text{rel}} n N_{\text{A}}}{V} = \sigma v_{\text{rel}} N_{\text{A}} \frac{n}{V}$$

For constant T, $v_{\text{rel}} = (16RT/\pi M)^{1/2}$ is also constant.

Since $pV = nRT$,

$$z = \sigma v_{\text{rel}} N_{\text{A}} \frac{p}{RT} = \sigma v_{\text{rel}} \frac{p}{k_{\text{B}} T} \propto p \quad \text{(at constant } T\text{)}$$

Brief Illustration 1B.1(b)

What is the collision frequency of an N_2 molecule in a sample at 1.00 atm and 25 °C? (σ = 0.43 nm^2)

From Brief Illustration 1B.1, v_{rel} = 671 m s^{-1}.

$$z = \frac{\sigma v_{\mathrm{rel}} p}{k_{\mathrm{B}} T} = \frac{(0.43 \times 10^{-18}\ \mathrm{m^2}) \times (671\ \mathrm{m\ s^{-1}}) \times (1.01 \times 10^{5}\ \mathrm{Pa})}{(1.381 \times 10^{-23}\ \mathrm{J\ K^{-1}}) \times (298\ \mathrm{K})}$$

$$z = 7.1 \times 10^{9}\ \mathrm{s^{-1}}$$

T-dependences of z

$$z = \frac{\sigma v_{\text{rel}} n N_{\text{A}}}{V} = \sigma v_{\text{rel}} N_{\text{A}} \frac{n}{V}$$

Since $v_{\text{rel}} = (16RT/\pi M)^{1/2}$,

$$z = \sigma N_{\text{A}} \frac{n}{V} \left(\frac{16RT}{\pi M}\right)^{1/2} \propto \sqrt{T} \quad \text{(at constant } V\text{)}$$

Since $pV = nRT$,

$$z = \sigma N_{\text{A}} \frac{p}{RT} \left(\frac{16RT}{\pi M}\right)^{1/2} \propto \frac{1}{\sqrt{T}} \quad \text{(at constant } p\text{)}$$

Mean Free Path, λ

"Average distance a molecule travels between collisions"

$$\lambda = \text{speed} \times \text{time} = v_{\mathrm{rel}} \times (1/z) = v_{\mathrm{rel}}/z$$

Hence,

$$\lambda = \frac{k_B T}{\sigma p}$$

Brief Illustration 1B.2

What is the mean free path of an N_2 molecule in a sample at 1.00 atm and 25 °C?

From Brief Illustration 1B.1(a), v_{rel} = 671 m s^{-1}.

From Brief Illustration 1B.1(b), z = 7.1 × 10^{9} s^{-1}.

$$\lambda = \frac{v_{\mathrm{rel}}}{z} = \frac{671\ \mathrm{m\ s^{-1}}}{7.1 \times 10^{9}\ \mathrm{s^{-1}}} = 9.5 \times 10^{-8}\ \mathrm{m}$$

Prelude to Chemical Kinetics

$$z = \sigma v_{\mathrm{rel}} N_{\mathrm{A}} \frac{n}{V}$$

Suppose that we have two types of molecules, A and B.

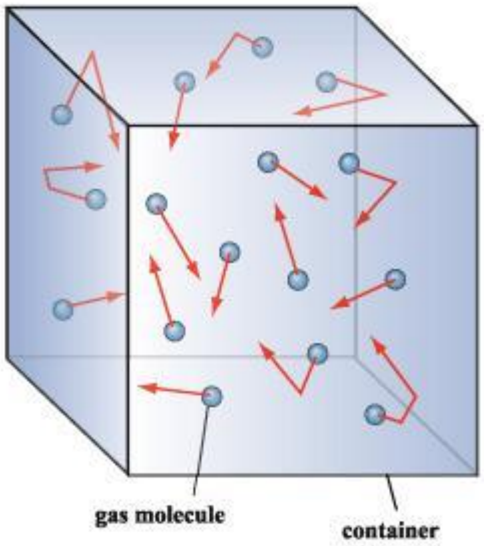
The collision rate of one A molecule with B molecules is

$$z = \sigma v_{\mathrm{rel}} N_{\mathrm{A}} \frac{n_{\mathrm{B}}}{V}$$

Hence, the collision density $Z_{\mathrm{AB}} = z(N_{\mathrm{A}} n_{\mathrm{A}}/V)$

$$Z_{\mathrm{AB}} = \sigma v_{\mathrm{rel}} N_{\mathrm{A}}^2 \frac{n_{\mathrm{A}}}{V} \frac{n_{\mathrm{B}}}{V} = \sigma \left(\frac{8 k_B T}{\pi \mu} \right)^{1/2} N_{\mathrm{A}}^2 [\mathrm{A}][\mathrm{B}] \propto [\mathrm{A}][\mathrm{B}]$$

Molecular Origin of Pressure



(Image: http://web2.uwindsor.ca/courses/physics/high_schools/2005/Brownian_motion/kinetictheory.jpg)

Position and Velocity

Each particle with mass m has its own

- Position $\vec{r} = (x, y, z)$
- Velocity $\vec{v} = (v_x, v_y, v_z)$

$$\vec{v} = \frac{\mathrm{d}\vec{r}}{\mathrm{d}t} = \left(\frac{\mathrm{d}x}{\mathrm{d}t}, \frac{\mathrm{d}y}{\mathrm{d}t}, \frac{\mathrm{d}z}{\mathrm{d}t}\right)$$

We can define linear momentum $\vec{p} = m\vec{v}$

Review

Newton's Second Law of Motion

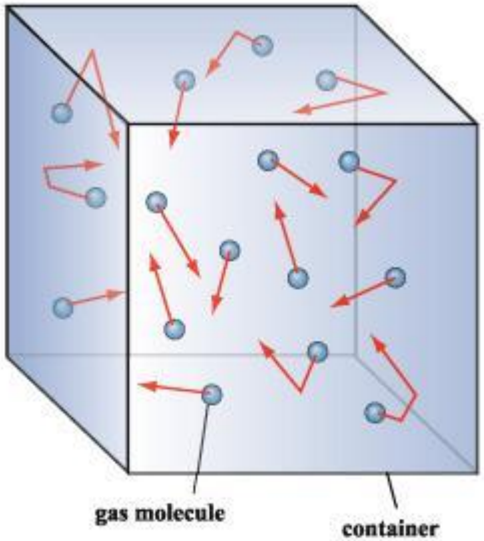
Acceleration $\vec{a} = \frac{\mathrm{d}\vec{v}}{\mathrm{d}t}$

Acceleration is determined by the force $\vec{F}$

$$\vec{F} = m\vec{a} = m\frac{\mathrm{d}\vec{v}}{\mathrm{d}t} = \frac{\mathrm{d}\vec{p}}{\mathrm{d}t}$$

If $\vec{F} = 0, \frac{\mathrm{d}\vec{p}}{\mathrm{d}t} = 0$ (momentum conservation)

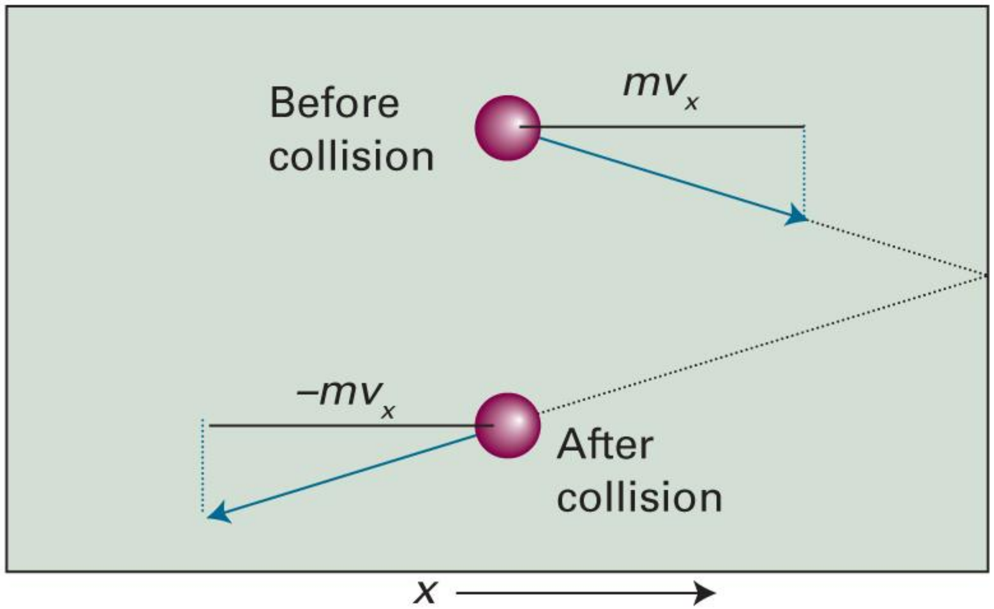
Derivation of Pressure



- Pressure $p = F/A$
- Force $\vec{F} = \mathrm{d}\vec{p}/\mathrm{d}t$: from collisions against the wall

(Image: http://web2.uwindsor.ca/courses/physics/high_schools/2005/Brownian_motion/kinetictheory.jpg)

The Change in Momentum



Change in momentum: $mv_x \rightarrow -mv_x$

$$\Delta p_{\text{particle}} = -mv_x - mv_x = -2mv_x$$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 1B.1)

Momentum Change of the Wall

- The particle has the momentum change of

$$\Delta p_{\text{particle}} = -mv_x - mv_x = -2mv_x$$

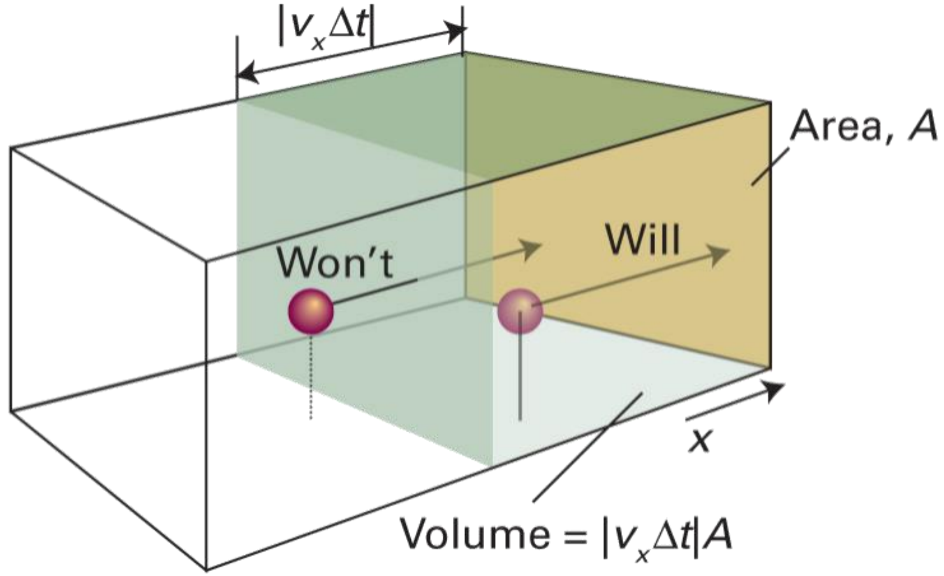
- The wall has the momentum change of

$$\Delta p_{\text{wall}} = 2mv_x$$

How Many Molecules Collide

Assume that all molecules have the same velocity.

For time Δt,



Number of molecules = volume × density

$$= (Av_x\Delta t) \times (N_{\mathrm{A}}n/V)$$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 1B.2)

Total Momentum Change

- For each molecule colliding the wall, the wall obtains

$$\Delta p_{\text{wall}} = 2mv_x$$

- Number of molecules = $(Av_x\Delta t) \times (N_A n/V)$
- 50% moving to the right, 50% to the left

- Total momentum change

$$\Delta p_{\text{wall,total}} = (2mv_x) \times (Av_x\Delta t)\left(\frac{N_A n}{V}\right) \times \frac{1}{2} = \frac{n\overbrace{mN_{\text{A}}}^{M}Av_x^2\Delta t}{V}$$

Force and Pressure

- Total momentum change $\Delta p_{\text{wall,total}} = nMAv_x^2\Delta t/V$
- Force exerted on the wall

$$F = \frac{\Delta p_{\text{wall,total}}}{\Delta t} = \frac{nMAv_x^2}{V}$$

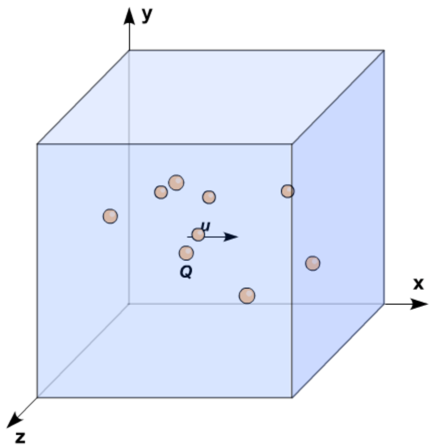
- Pressure exerted on the wall

$$p = \frac{F}{A} = \frac{nMv_x^2}{V}$$

(under the same velocity assumption)

Same Velocity Assumption

- In reality, each particle has a different velocity



- Replace the value by the mean value:

$$v_x^2 \rightarrow \langle v_x^2 \rangle$$

(Image: https://www.pinterest.ca/pin/314970567674652846/)

Pressure

$$p = \frac{nMv_x^2}{V} \quad \rightarrow \quad p = \frac{nM\langle v_x^2 \rangle}{V}$$

- Total velocity $v^2 = v_x^2 + v_y^2 + v_z^2$

$$\langle v^2 \rangle = \langle v_x^2 \rangle + \left\langle v_y^2 \right\rangle + \langle v_z^2 \rangle$$
$$= \langle v_x^2 \rangle + \langle v_x^2 \rangle + \langle v_x^2 \rangle = 3\langle v_x^2 \rangle$$

- Final formula

$$p = \frac{nM\langle v^2 \rangle}{3V}$$

p-T Relationship

$$pV = \frac{1}{3} nM\langle v^2 \rangle$$

From the Maxwell-Boltzmann distribution,

$$\langle v^2 \rangle = v_{\mathrm{rms}}^2 = \frac{3RT}{M}$$

Hence,

$$pV = nRT$$